STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-SMOOTHING-ARTINIAN-LCI

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3260-3266 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 3268-3290, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 27 of queue_examples-defos.json (pb=1070, stmt 3260-3266, proof 3268-3290).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00PL: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3290–3290. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00JB: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3290–3290. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04GG: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3290–3290. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04GM: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3290–3290. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00J6: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3290–3290. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00SA: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3290–3290. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7V: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3216–3221. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let k be a field. Let A be a finite dimensional k-algebra which is a local complete intersection over k. Then there is a finite flat $k[[t]]$-algebra B with $B/tB \cong A$ and $B[1/t]$ étale over $k((t))$.*

Proof. Since A is Artinian (Algebra, Lemma 00J6), we can write A as a product of local Artinian rings (Algebra, Lemma 00JB). Thus it suffices to prove the lemma if A is local (this uses that being a local complete intersection is preserved under taking principal localizations, see Algebra, Lemma 00SA). In this case A is a global complete intersection. Consider the algebra B constructed in Lemma 0E7V. Then $k[[t]] \to B$ is quasi-finite at the unique prime of B lying over (t) (Algebra, Definition 00PL). Observe that $k[[t]]$ is a henselian local ring (Algebra, Lemma 04GM). Thus $B = B' \times C$ where B' is finite over $k[[t]]$ and C has no prime lying over (t), see Algebra, Lemma 04GG. Then B' is the ring we are looking for (recall that étale is the same thing as smooth of relative dimension 0). □